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**Master of Computer Applications
Semester – III
Data Analytics for Business
Experiment No.: 08**

Title: Marketing campaign performance dashboard

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Experiment No.: 08

Marketing campaign performance dashboard

Problem Statement

You are a Marketing Analyst in a digital marketing agency. Multiple campaigns have been executed across different channels. Management wants to evaluate campaign performance based on impressions, clicks, conversions, and ROI. Develop a dashboard that provides meaningful campaign insights

Methodology / Steps Followed

Step 1: Data Collection

- Obtained Marketing Campaign data from Kaggle datasets.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the **Get Data** option.
- Loaded CSV file into Power BI.

Step 3: Data Cleaning and Transformation

- Removed duplicate records.
- Handled missing values.
- Converted data types where necessary

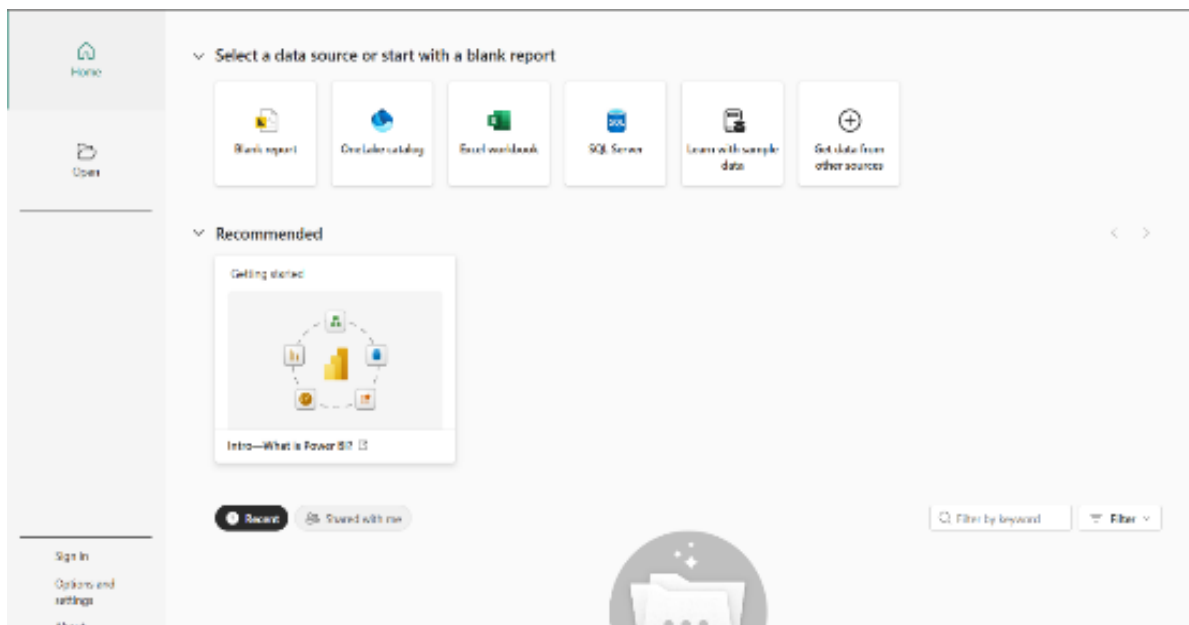
Step 4: Dashboard Development

- Created KPI Cards to display Total Campaigns, Total Impressions, Total Clicks, Average Conversion Rate, and Average ROI.
- Developed a Bar Chart to compare ROI across different marketing channels such as Email, Google Ads, Website, YouTube, Instagram, and Facebook.
- Developed a Line Chart to visualize the trend of Average ROI over time based on year, quarter, month, and day.
- Created a Column Chart to analyse Average ROI across different Campaign Types.
- Created a Donut Chart to analyse the distribution of campaigns across different Customer Segments.
- Developed a Bar Chart to compare Average Conversion Rate across different Locations.
- Created a Column Chart to compare Average Acquisition Cost across different Campaign Types.
- Developed a Bar Chart to analyse Engagement Score across different Marketing Channels.
- Created a Treemap to analyse the distribution of campaigns across different Companies.

- Developed a Donut Chart to analyse the distribution of campaigns across different Languages.
- Added Interactive Slicers for Campaign Type, Channel Used, Customer Segment, Target Audience, Acquisition Cost, Engagement Score, and Company to enable dynamic campaign performance analysis.

Output

- Opened Power BI file.



- Loaded the dataset using Get Data /CSV.
- Transform The data.
- Applied cleaning using power Query And Applied cleaning steps where ever necessary.

Marketing_Campaign

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns Keep Remove Rows Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Merge Queries Append Queries Combine Files Combine

Queries [1]

marketing_campaign_da... = Table.TransformColumnTypes(#"Removed Duplicates",{{"Campaign_ID", Int64.Type}, {"Company", type text}, {"Campaign_Type", type text}, {"Target_Audience", type text}, {"Duration", type text}, {"Channel_Used", type text}, {"Conversion_Rate", type decimal}}

Campaign_ID	Company	Campaign_Type	Target_Audience	Duration	Channel_Used	Conversion_Rate
1	Innovate Industries	Email	Men 18-24	30 days	Google Ads	
2	NexGen Systems	Email	Women 35-44	60 days	Google Ads	
3	Alpha Innovations	Influencer	Men 25-34	30 days	YouTube	
4	DataTech Solutions	Display	All Ages	60 days	YouTube	
5	NexGen Systems	Email	Men 25-34	15 days	YouTube	
6	DataTech Solutions	Display	All Ages	15 days	Instagram	
7	NexGen Systems	Email	Women 35-44	60 days	Website	
8	DataTech Solutions	Search	Men 18-24	45 days	Google Ads	
9	Alpha Innovations	Social Media	Women 35-44	15 days	Facebook	
10	TechCorp	Email	Women 35-44	15 days	Instagram	
11	NexGen Systems	Display	Men 25-34	45 days	Email	
12	Innovate Industries	Influencer	Men 25-34	60 days	Google Ads	
13	TechCorp	Social Media	Men 25-34	60 days	Facebook	
14	TechCorp	Email	Men 25-34	45 days	Instagram	
15	TechCorp	Display	All Ages	45 days	Website	
16	Innovate Industries	Social Media	Women 35-44	60 days	YouTube	
17	Innovate Industries	Display	Women 35-44	45 days	Website	
18	Alpha Innovations	Influencer	Men 18-24	15 days	Instagram	
19	Alpha Innovations	Social Media	Men 25-34	60 days	Google Ads	
20	DataTech Solutions	Influencer	Men 25-34	15 days	Google Ads	
21	DataTech Solutions	Search	Women 25-34	15 days	Email	

Query Settings

Properties

Name: marketing_campaign_dataset

Applied Steps

- Source
- Promoted headers
- Changed column type
- Removed Duplicates
- Changed Type

- Added Dax Queries to the KPI Cards

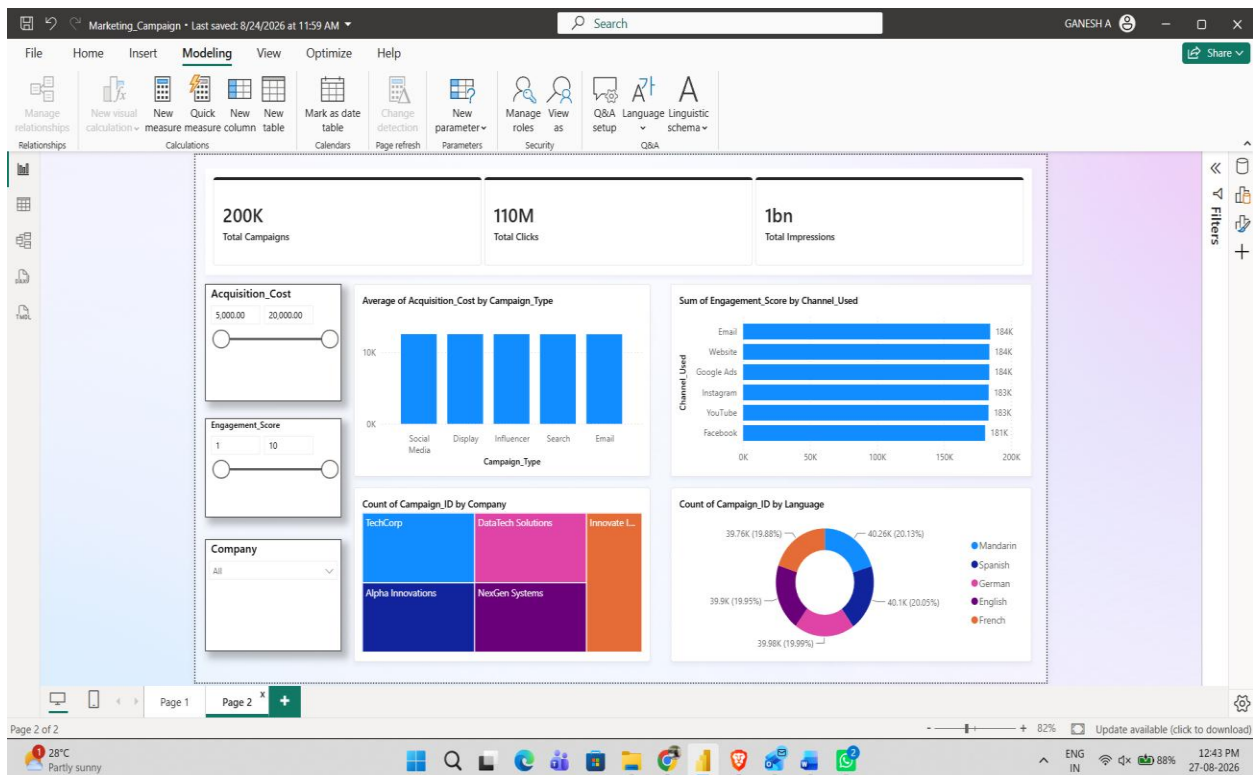
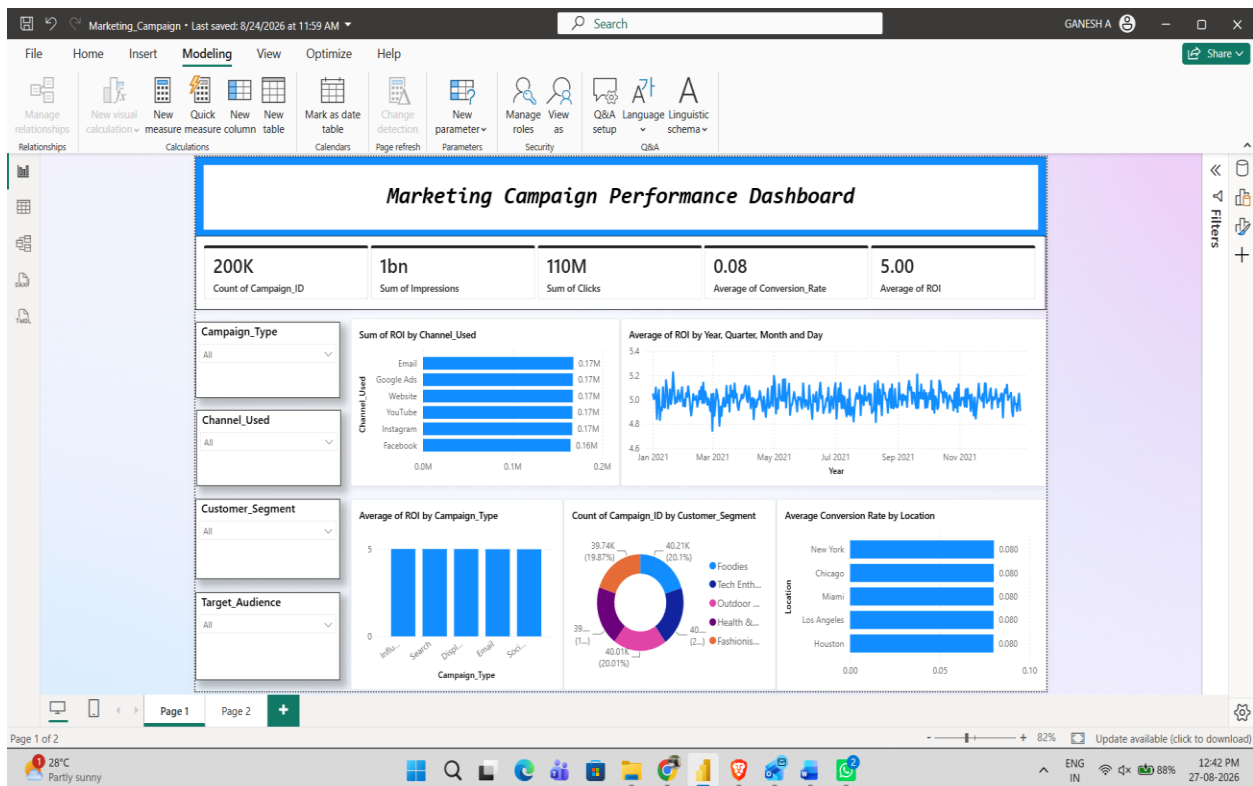
- Average Conversion Rate

AVERAGE(marketing_campaign_dataset[Conversion_Rate])

- Total Campaigns =

DISTINCTCOUNT('marketing_campaign_dataset'[Campaign_ID])

- Total Clicks = SUM(marketing_campaign_dataset[Clicks])



Result

- The dashboard gives a clear picture of the overall marketing campaign performance, covering around 200K campaigns.
- The campaigns achieved nearly 1 billion impressions and 110 million clicks, showing that the marketing activities reached a large audience.
- The overall average conversion rate is 0.08, which provides an indication of how many users responded positively to the campaigns.
- The average ROI is around 5, suggesting that the campaigns generated a good return compared with the investment made.
- The performance of different marketing channels was quite similar, with Email, Google Ads, Website, YouTube, Instagram, and Facebook showing comparable ROI values.
- The ROI trend remained mostly stable throughout 2021, although there were small changes from month to month.
- The different campaign types recorded nearly similar ROI values, showing that no particular campaign type had a major advantage in terms of return.
- The number of campaigns was fairly evenly distributed among customer segments, providing a balanced view of audience targeting.
- The conversion rate across locations was almost the same, indicating consistent campaign performance in the selected cities.
- Acquisition costs were also relatively close across campaign types, showing that campaign spending did not vary significantly.
- The dashboard shows differences in engagement levels across channels, which can help identify channels that attract better audience interaction.
- Campaigns were distributed across different companies and languages, providing additional insights into how marketing activities were spread.
- Overall, the dashboard helps convert large amounts of campaign data into easy-to-understand insights, making it useful for comparing campaigns and supporting better marketing decisions.

Conclusion

The Marketing Campaign Performance Dashboard provides a simple and effective way to understand the overall performance of different marketing activities. By analysing campaigns across channels, campaign types, customer segments, locations, companies, and languages, the dashboard gives a clear view of important factors such as impressions, clicks, conversion rate, engagement, acquisition cost, and ROI. The results show that the campaigns achieved strong reach and generated positive returns, while performance remained relatively consistent across most categories. The interactive filters also make it easier to explore specific areas and compare results. Overall, the dashboard helps transform campaign data into meaningful insights that can support better planning, performance evaluation, and data-driven marketing decisions.

Learning Outcomes

Through this project, I learned:

- Learned how to analyse marketing campaign data and identify important performance indicators.
- Gained practical experience in Power BI dashboard development.
- Learned to create and use KPI cards for presenting key campaign metrics.
- Developed skills in selecting suitable charts and visualizations for different types of data.
- Learned how to compare campaign performance across channels, campaign types, locations, and customer segments.
- Gained an understanding of important marketing metrics such as ROI, conversion rate, impressions, clicks, engagement, and acquisition cost.
- Learned how to use interactive slicers and filters to explore data from different perspectives.
- Improved the ability to identify trends, patterns, and differences in campaign performance.
- Developed skills in presenting complex data in a simple and visually understandable format.
- Gained experience in using data visualization to support data-driven marketing decisions.

Github Link:

https://github.com/GANESH-A08/Data_Analytics_projects/tree/main/Week%208_Experiment